2.50 Coupled 1st Order Linear DEs

Worked Solutions

1) a) $\dfrac{dy}{dt} = y$. Use separation of variables:

$$\int \frac{1}{y}\,dy = \int 1\,dt \quad \Longrightarrow \quad \ln|y| = t + c \quad \Longrightarrow \quad y = Ae^t.$$

1) b) Using $y = Ae^t$,

$$\frac{dx}{dt} = x + Ae^t \quad \Longrightarrow \quad \frac{dx}{dt} - x = Ae^t.$$

The integrating factor is $e^{\int -1\,dt} = e^{-t}$. Hence

$$e^{-t}\frac{dx}{dt} - xe^{-t} = A, \quad \frac{d}{dt}(xe^{-t}) = A, \quad xe^{-t} = At + c,$$

so $x = (At + c)e^t$.

2) a) $\dfrac{dx}{dt} = 2x$. By separation of variables,

$$\int \frac{1}{x}\,dx = \int 2\,dt \quad \Longrightarrow \quad \ln|x| = 2t + c \quad \Longrightarrow \quad x = Ae^{2t}.$$

2) b) Using $x = Ae^{2t}$,

$$\frac{dy}{dt} = Ae^{2t} + 2y \quad \Longrightarrow \quad \frac{dy}{dt} - 2y = Ae^{2t}.$$

The integrating factor is $e^{\int -2\,dt} = e^{-2t}$. Hence

$$e^{-2t}\frac{dy}{dt} - 2ye^{-2t} = A, \quad \frac{d}{dt}(ye^{-2t}) = A, \quad ye^{-2t} = At + c,$$

so $y = (At + c)e^{2t}$.

3) a) From $\dfrac{dx}{dt} = x + y$, $y = \dfrac{dx}{dt} - x$.

3) b) Differentiating,

$$\frac{dy}{dt} = \frac{d^2x}{dt^2} - \frac{dx}{dt}.$$

3) c) Substituting into $\dfrac{dy}{dt} = 2y$ gives

$$\frac{d^2x}{dt^2} - \frac{dx}{dt} = 2\left(\frac{dx}{dt} - x\right) \quad \Longrightarrow \quad x'' - 3x' + 2x = 0.$$

3) d) The auxiliary equation is $m^2 - 3m + 2 = 0 = (m-1)(m-2)$, so $m = 1$ or $m = 2$. Thus $x = Ae^t + Be^{2t}$.

3) e) Since $y = x' - x$,

$$y = Ae^t + 2Be^{2t} - (Ae^t + Be^{2t}) = Be^{2t}.$$

4) a) From $x' = 2x + y$, $y = x' - 2x$, so $y' = x'' - 2x'$. Using $y' = x + 2y$,

$$x'' - 2x' = x + 2(x' - 2x) \quad \Longrightarrow \quad x'' - 4x' + 3x = 0.$$

4) b) The auxiliary equation $m^2 - 4m + 3 = 0 = (m-1)(m-3)$ gives $m = 1$ or $m = 3$. Hence $x = Ae^t + Be^{3t}$.

4) c) Since $y = x' - 2x$,

$$y = Ae^t + 3Be^{3t} - 2(Ae^t + Be^{3t}) = -Ae^t + Be^{3t}.$$

5) a) From $y' = -4x + 5y$, $x = \frac{5}{4}y - \frac{1}{4}y'$, so $x' = \frac{5}{4}y' - \frac{1}{4}y''$. Using $x' = x + y$,

$$\frac{5}{4}y' - \frac{1}{4}y'' = \left(\frac{5}{4}y - \frac{1}{4}y'\right) + y = \frac{9}{4}y - \frac{1}{4}y',$$
$$y'' - 6y' + 9y = 0.$$

5) b) The auxiliary equation is $m^2 - 6m + 9 = 0 = (m-3)^2$, so $m = 3$. Thus $y = Ate^{3t} + Be^{3t}$.

5) c) Using $x = \frac{5}{4}y - \frac{1}{4}y'$ and $y' = 3Ate^{3t} + Ae^{3t} + 3Be^{3t}$,

$$\begin{aligned} x &= \frac{5}{4}(Ate^{3t} + Be^{3t}) - \frac{1}{4}(3Ate^{3t} + Ae^{3t} + 3Be^{3t}) \\ &= \frac{1}{2}Ate^{3t} - \frac{1}{4}Ae^{3t} + \frac{1}{2}Be^{3t}. \end{aligned}$$

6) a) Since $x' = y$, $y' = x''$. Using $y' = -x + 2y$ gives $x'' = -x + 2x'$, hence $x'' - 2x' + x = 0$.

6) b) The auxiliary equation $m^2 - 2m + 1 = 0 = (m-1)^2$ gives $x = Ate^t + Be^t$. Therefore $x' = Ate^t + Ae^t + Be^t$, and $y = x' = Ate^t + Ae^t + Be^t$.

6) c) From $x(0) = 1$, $B = 1$. From $y(0) = 0$, $A + 1 = 0$, so $A = -1$. Thus $x = -te^t + e^t$ and $y = -te^t$.

7) a) From $y' = x - y$, $x = y' + y$, so $x' = y'' + y'$. Using $x' = x + 3y$,

$$y'' + y' = (y' + y) + 3y \quad \Longrightarrow \quad y'' - 4y = 0.$$

7) b) The auxiliary equation $m^2 - 4 = 0$ gives $m = 2$ or $m = -2$, so $y = Ae^{2t} + Be^{-2t}$.

7) c) Since $x = y' + y$,

$$x = (2Ae^{2t} - 2Be^{-2t}) + (Ae^{2t} + Be^{-2t}) = 3Ae^{2t} - Be^{-2t}.$$

7) d) Using $x(0) = 4$ and $y(0) = 0$,

$$3A - B = 4, \qquad A + B = 0.$$

Solving gives $A = 1$ and $B = -1$. Hence $x = 3e^{2t} + e^{-2t}$ and $y = e^{2t} - e^{-2t}$.

8) a) From $x' = 2x + y$, $y = x' - 2x$ and $y' = x'' - 2x'$. Using $y' = x + 2y$ gives $x'' - 4x' + 3x = 0$. Its auxiliary equation $(m-1)(m-3) = 0$ gives $x = Ae^t + Be^{3t}$. Then $y = x' - 2x = -Ae^t + Be^{3t}$. Thus $x = Ae^t + Be^{3t}$ and $y = -Ae^t + Be^{3t}$.

8) b) From $x' = 2x + 2y$, $y = \frac{1}{2}x' - x$ and $y' = \frac{1}{2}x'' - x'$. Using $y' = -2x + 6y$ gives $x'' - 8x' + 16x = 0$. The auxiliary equation $(m-4)^2 = 0$ gives $x = Ate^{4t} + Be^{4t}$. Then

$$y = \tfrac{1}{2}x' - x = Ate^{4t} + \tfrac{1}{2}Ae^{4t} + Be^{4t}.$$

8) c) From $x' = x + 2y$, $y = \frac{1}{2}x' - \frac{1}{2}x$. Using $y' = -x - y$ gives $x'' + x = 0$. Thus $x = A\cos t + B\sin t$, and

$$y = \tfrac{1}{2}x' - \tfrac{1}{2}x = -\tfrac{1}{2}(A + B)\sin t + \tfrac{1}{2}(-A + B)\cos t.$$

8) d) From $x' = 2x + y$, $y = x' - 2x$. Using $y' = -5x$ gives $x'' - 2x' + 5x = 0$. The auxiliary equation gives $m = 1 \pm 2i$, so $x = e^t(A\cos 2t + B\sin 2t)$. Then $y = x' - 2x$, hence

$$y = (-2A - B)e^t \sin 2t + (-A + 2B)e^t \cos 2t.$$

9) a) From $x' = x + y + 1$, $y = x' - x - 1$, so $y' = x'' - x'$. Using $y' = x + y - 1$ gives

$$x'' - x' = x + (x' - x - 1) - 1 \quad \Longrightarrow \quad x'' - 2x' = -2.$$

9) b) The homogeneous equation is $x'' - 2x' = 0$, with auxiliary equation $m(m - 2) = 0$. Thus the complementary function is $x = Ae^{2t} + B$. Try $x = pt$ for a particular integral; then $x' = p$, $x'' = 0$, and $-2p = -2$, so $p = 1$. Hence $x = Ae^{2t} + B + t$.

9) c) Since $y = x' - x - 1$,

$$y = (2Ae^{2t} + 1) - (Ae^{2t} + B + t) - 1 = Ae^{2t} - B - t.$$

9) d) Using $x(0) = 1$ and $y(0) = 0$ gives $A + B = 1$ and $A - B = 0$. Thus $A = B = \frac{1}{2}$, and

$$x = \tfrac{1}{2}e^{2t} + \tfrac{1}{2} + t, \qquad y = \tfrac{1}{2}e^{2t} - \tfrac{1}{2} - t.$$

10) a) From $x' = 2y + 6$, $y = \frac{1}{2}x' - 3$, so $y' = \frac{1}{2}x''$. Using $y' = 2x - 4$ gives $x'' - 4x = -8$. The complementary function is $Ae^{2t} + Be^{-2t}$. A constant particular integral $x = k$ gives $-4k = -8$, so $k = 2$. Therefore

$$x = Ae^{2t} + Be^{-2t} + 2, \qquad y = \tfrac{1}{2}x' - 3 = Ae^{2t} - Be^{-2t} - 3.$$

10) b) From $x' = 3x+y+2$, $y = x'-3x-2$ and $y' = x''-3x'$. Using $y' = -x+y-2$ gives $x''-4x'+4x = -4$. The complementary function is $Ate^{2t}+Be^{2t}$. A constant particular integral gives $4k = -4$, so $k = -1$. Hence

$$x = Ate^{2t} + Be^{2t} - 1,$$

and $y = x' - 3x - 2 = -Ate^{2t} + (A - B)e^{2t} + 1$.

10) c) From $x' = 2x + y + 3$, $y = x' - 2x - 3$ and $y' = x'' - 2x'$. Substitution in $y' = -13x - 2y - 5$ gives $x'' + 9x = 1$. Its complementary function is $A\cos 3t + B\sin 3t$; the constant particular integral is $1/9$. Thus

$$x = A\cos 3t + B\sin 3t + \tfrac{1}{9}.$$

Then $y = x' - 2x - 3$, so

$$y = (-3A - 2B)\sin 3t + (3B - 2A)\cos 3t - \tfrac{29}{9}.$$

10) d) From $x' = 3x + 2y + 1$, $y = \frac{1}{2}x' - \frac{3}{2}x - \frac{1}{2}$ and $y' = \frac{1}{2}x'' - \frac{3}{2}x'$. Substitution in $y' = -2x + 3y + 1$ gives

$$x'' - 6x' + 13x = -1.$$

The auxiliary equation gives $m = 3 \pm 2i$, so the complementary function is $e^{3t}(A\cos 2t + B\sin 2t)$. A constant particular integral gives $13k = -1$, hence $k = -1/13$. Therefore

$$x = e^{3t}(A\cos 2t + B\sin 2t) - \tfrac{1}{13}.$$

Using $y = \frac{1}{2}x' - \frac{3}{2}x - \frac{1}{2}$ gives

$$y = e^{3t}(B\cos 2t - A\sin 2t) - \tfrac{5}{13}.$$

11) F) rom $x' = x + y + t$, $y = x' - x - t$, so $y' = x'' - x' - 1$. Using $y' = 2y + t^2$ gives

$$x'' - 3x' + 2x = 1 - 2t + t^2.$$

The complementary function is $Ae^t + Be^{2t}$. Try $x = pt^2 + qt + r$. Comparing coefficients in the differential equation gives

$$2p = 1, \quad -6p + 2q = -2, \quad 2p - 3q + 2r = 1,$$

so $p = q = \frac{1}{2}$ and $r = \frac{3}{4}$. Thus

$$x = Ae^t + Be^{2t} + \tfrac{1}{2}t^2 + \tfrac{1}{2}t + \tfrac{3}{4}, \quad y = x' - x - t = Be^{2t} - \tfrac{1}{2}t^2 - \tfrac{1}{2}t - \tfrac{1}{4}.$$

Using $x(0) = 7/4$ and $y(0) = 1/4$ gives $A + B = 1$ and $B = 1/2$, so $A = 1/2$. Hence

$$x = \tfrac{1}{2}e^t + \tfrac{1}{2}e^{2t} + \tfrac{1}{2}t^2 + \tfrac{1}{2}t + \tfrac{3}{4}, \quad y = \tfrac{1}{2}e^{2t} - \tfrac{1}{2}t^2 - \tfrac{1}{2}t - \tfrac{1}{4}.$$

12) a) From $x' = 2x + y + \sin t$, $y = x' - 2x - \sin t$, so $y' = x'' - 2x' - \cos t$. Using $y' = -x + \cos t$ gives

$$x'' - 2x' + x = 2\cos t.$$

12) b) The complementary function is $Ate^t + Be^t$. Try $x = p\cos t + q\sin t$. Substitution gives $2p\sin t - 2q\cos t = 2\cos t$, so $p = 0$, $q = -1$. Hence $x = Ate^t + Be^t - \sin t$.

12) c) Since $y = x' - 2x - \sin t$,

$$y = -Ate^t + (A - B)e^t - \cos t + \sin t.$$

12) d) Using $x(0) = 1$ gives $B = 1$. Using $y(0) = 1$ gives $A = 3$. Therefore

$$x = 3te^t + e^t - \sin t, \qquad y = -3te^t + 2e^t - \cos t + \sin t.$$

13) a) From $x' = 2x+y+\sin t$, $y = x'-2x-\sin t$ and $y' = x''-2x'-\cos t$. Substitution in $y' = 2x+3y+\cos t$ gives

$$x'' - 5x' + 4x = -3\sin t + 2\cos t.$$

The complementary function is $Ae^t + Be^{4t}$. With trial integral $p\cos t + q\sin t$, coefficient comparison gives $3p - 5q = 2$ and $5p + 3q = -3$, so $p = -9/34$, $q = -19/34$. Thus

$$x = Ae^t + Be^{4t} - \tfrac{9}{34}\cos t - \tfrac{19}{34}\sin t.$$

Using $y = x' - 2x - \sin t$ gives

$$y = -Ae^t + 2Be^{4t} + \tfrac{13}{34}\sin t - \tfrac{1}{34}\cos t.$$

13) b) From $x' = x + y + \sin 4t$, $y = x' - x - \sin 4t$. Substitution in $y' = -x + 3y + 3\sin 4t$ gives

$$x'' - 4x' + 4x = 4\cos 4t.$$

The complementary function is $Ate^{2t} + Be^{2t}$. With trial integral $p\cos 4t + q\sin 4t$, coefficient comparison gives $-12p - 16q = 4$ and $16p - 12q = 0$, so $p = -3/25$, $q = -4/25$. Therefore

$$x = Ate^{2t} + Be^{2t} - \tfrac{3}{25}\cos 4t - \tfrac{4}{25}\sin 4t.$$

Using $y = x' - x - \sin 4t$ gives

$$y = Ate^{2t} + (A + B)e^{2t} - \tfrac{13}{25}\cos 4t - \tfrac{9}{25}\sin 4t.$$

13) c) From $x' = 3x + 2y + \cos 2t$, $y = \frac{1}{2}x' - \frac{3}{2}x - \frac{1}{2}\cos 2t$ and $y' = \frac{1}{2}x'' - \frac{3}{2}x' + \sin 2t$. Substitution in $y' = -9x - 3y - 5\cos 2t$ gives

$$x'' + 9x = -7\cos 2t - 2\sin 2t.$$

The complementary function is $A\cos 3t + B\sin 3t$. For a trial integral $p\cos 2t + q\sin 2t$, coefficient comparison gives $5p = -7$ and $5q = -2$, so $p = -7/5$ and $q = -2/5$. Therefore

$$x = A\cos 3t + B\sin 3t - \tfrac{7}{5}\cos 2t - \tfrac{2}{5}\sin 2t.$$

Using $y = \frac{1}{2}x' - \frac{3}{2}x - \frac{1}{2}\cos 2t$ gives

$$y = \tfrac{3}{2}(-A + B)\cos 3t - \tfrac{3}{2}(A + B)\sin 3t + \tfrac{6}{5}\cos 2t + 2\sin 2t.$$

13) d) From $x' = 2x + y + \sin 5t$, $y = x' - 2x - \sin 5t$. Substitution in $y' = -x + 2y - 25\cos 5t + 22\sin 5t$ gives

$$x'' - 4x' + 5x = -20\cos 5t + 20\sin 5t.$$

The complementary function is $e^{2t}(A\cos t + B\sin t)$. For a trial integral $p\cos 5t + q\sin 5t$, coefficient comparison gives $-20p - 20q = -20$ and $20p - 20q = 20$, so $p = 1$ and $q = 0$. Hence

$$x = e^{2t}(A\cos t + B\sin t) + \cos 5t.$$

Using $y = x' - 2x - \sin 5t$ gives

$$y = e^{2t}(B\cos t - A\sin t) - 2\cos 5t - 6\sin 5t.$$

14) F) rom $x' = 2x + 3y + e^{-2t}$,

$$y = \tfrac{1}{3}x' - \tfrac{2}{3}x - \tfrac{1}{3}e^{-2t}, \qquad y' = \tfrac{1}{3}x'' - \tfrac{2}{3}x' + \tfrac{2}{3}e^{-2t}.$$

Using $y' = x + 2y + e^{-2t}$ gives $x'' - 4x' + x = -e^{-2t}$. The auxiliary equation $m^2 - 4m + 1 = 0$ gives $m = 2 \pm \sqrt{3}$, so the complementary function is $Ae^{(2+\sqrt{3})t} + Be^{(2-\sqrt{3})t}$. A trial integral ke^{-2t} gives $13k = -1$, hence $k = -1/13$. Thus

$$x = Ae^{(2+\sqrt{3})t} + Be^{(2-\sqrt{3})t} - \tfrac{1}{13}e^{-2t},$$

$$y = \tfrac{\sqrt{3}}{3}Ae^{(2+\sqrt{3})t} - \tfrac{\sqrt{3}}{3}Be^{(2-\sqrt{3})t} - \tfrac{3}{13}e^{-2t}.$$

Using $x(0) = 25/13$ and $y(0) = -3/13$ gives $A + B = 2$ and $A - B = 0$, so $A = B = 1$. Therefore

$$x = e^{(2+\sqrt{3})t} + e^{(2-\sqrt{3})t} - \tfrac{1}{13}e^{-2t}, \quad y = \tfrac{\sqrt{3}}{3}e^{(2+\sqrt{3})t} - \tfrac{\sqrt{3}}{3}e^{(2-\sqrt{3})t} - \tfrac{3}{13}e^{-2t}.$$

15) F) rom $x' = -3x + 4y + e^{-t}$,

$$y = \tfrac{1}{4}x' + \tfrac{3}{4}x - \tfrac{1}{4}e^{-t}, \qquad y' = \tfrac{1}{4}x'' + \tfrac{3}{4}x' + \tfrac{1}{4}e^{-t}.$$

Using $y' = \frac{3}{2}x - 4y + e^{-t}$ gives $x'' + 7x' + 6x = 7e^{-t}$. The auxiliary equation $(m + 1)(m + 6) = 0$ gives complementary function $Ae^{-t} + Be^{-6t}$. A trial integral kte^{-t} gives $5k = 7$, so $k = 7/5$. Hence

$$x = Ae^{-t} + Be^{-6t} + \tfrac{7}{5}te^{-t}.$$

Using $y = \frac{1}{4}x' + \frac{3}{4}x - \frac{1}{4}e^{-t}$ gives

$$y = \left(\tfrac{1}{2}A + \tfrac{1}{10}\right)e^{-t} - \tfrac{3}{4}Be^{-6t} + \tfrac{7}{10}te^{-t}.$$

The initial conditions $x(0) = 1$ and $y(0) = 1/4$ give $A + B = 1$ and $\frac{1}{2}A - \frac{3}{4}B = \frac{3}{20}$. Thus $A = 18/25$ and $B = 7/25$, so

$$x = \tfrac{18}{25}e^{-t} + \tfrac{7}{25}e^{-6t} + \tfrac{7}{5}te^{-t}, \quad y = \tfrac{23}{50}e^{-t} - \tfrac{21}{100}e^{-6t} + \tfrac{7}{10}te^{-t}.$$

16) a) The $0.3x$ term represents natural growth of the rabbit population. The $-12y$ term represents the decrease in rabbits due to predation by foxes.

16) b) The $0.02x$ term represents the increase in foxes due to the availability of rabbits as food. The -1.5 term represents a constant loss of foxes each year. As it does not involve x or y, foxes are lost at a fixed rate, regardless of how many foxes there are.

16) c) If there are no foxes, then $y = 0$ and $\dfrac{dx}{dt} = 0.3x$. Since $0.3x > 0$, the rabbit population would increase exponentially.

16) d) If there are no rabbits, then $x = 0$ and $\dfrac{dy}{dt} = -1.5$. Hence the fox population would decrease steadily at 1.5 foxes per year.

16) e) In a real population, deaths would usually depend on how many foxes there are. If there are more foxes, you would expect more deaths overall, and if there were few foxes then you would expect fewer deaths. In reality, deaths would usually be proportional to the fox population, so a term like $-1.5y$ would be more realistic.

17) a) The coefficient 0.5 of x in the dx/dt equation represents the natural growth rate of the fish population. The coefficient -8 of y in the dx/dt equation represents the decrease in fish due to predation by birds. The coefficient 0.02 of x in the dy/dt equation represents the increase in birds due to the availability of fish as food. The constant -1.6 in the dy/dt equation represents a fixed decrease in the bird population.

17) b) The equilibrium point exists when both derivatives are zero:

$$0.5x - 8y = 0, \qquad 0.02x - 1.6 = 0.$$

The second equation gives $x = 1.6/0.02 = 80$. Substitution into the first gives $0.5(80) - 8y = 0$, so $y = 5$. The populations remain constant if there are 80 fish and 5 birds.

18) a) Comparing the model with the assumptions gives $p = 40$, $q = 4$, $r = 2$, and $s = 2$.

18) b) From $x' = 0.4x - 4y$, $y = \frac{1}{10}x - \frac{1}{4}x'$, so $y' = \frac{1}{10}x' - \frac{1}{4}x''$. Using $y' = 0.02x - 2$ gives

$$x'' - \tfrac{2}{5}x' + \tfrac{2}{25}x = 8.$$

The auxiliary equation for the homogeneous equation is $m^2 - \frac{2}{5}m + \frac{2}{25} = 0$, so $m = \frac{1}{5} \pm \frac{1}{5}i$. The complementary function is $e^{t/5}(A\cos(t/5) + B\sin(t/5))$. A constant particular integral $x = k$ gives $\frac{2}{25}k = 8$, so $k = 100$. Thus

$$x = e^{t/5}\left(A\cos\tfrac{t}{5} + B\sin\tfrac{t}{5}\right) + 100.$$

Using $y = \frac{1}{10}x - \frac{1}{4}x'$ gives

$$y = e^{t/5}\left(\left(\tfrac{1}{20}A - \tfrac{1}{20}B\right)\cos\tfrac{t}{5} + \left(\tfrac{1}{20}A + \tfrac{1}{20}B\right)\sin\tfrac{t}{5}\right) + 10.$$

From $x(0) = 1200$, $A = 1100$. From $y(0) = 25$, $55 - \frac{1}{20}B + 10 = 25$, so $B = 800$. Therefore

$$x = e^{t/5}\left(1100\cos\tfrac{t}{5} + 800\sin\tfrac{t}{5}\right) + 100, \quad y = e^{t/5}\left(15\cos\tfrac{t}{5} + 95\sin\tfrac{t}{5}\right) + 10.$$

18) c) At $t = 4$,

$$x = e^{4/5}\left(1100\cos\tfrac{4}{5} + 800\sin\tfrac{4}{5}\right) + 100 = 3082.808\ldots \approx 3083.$$

The initial insect population is 1200, so it has increased; therefore the insect population is not being successfully controlled.

19) a) Comparing the assumptions with the model gives $A = -0.05$, $B = -2$, $C = 0.02$, and $D = -0.05$.

19) b) The owl death term being proportional to y means that owl deaths depend on the size of the owl population: the larger the owl population, the more owls die each year.

19) c) The system is

$$x' = -0.05x - 2y, \qquad y' = 0.02x - 0.05y.$$

From the first equation, $y = -\frac{1}{40}x - \frac{1}{2}x'$, so $y' = -\frac{1}{40}x' - \frac{1}{2}x''$. Substitution into the second equation gives

$$x'' + \tfrac{1}{10}x' + \tfrac{17}{400}x = 0.$$

The auxiliary equation gives $m = -\frac{1}{20} \pm \frac{1}{5}i$. Hence

$$x = e^{-t/20}\left(A\cos\tfrac{t}{5} + B\sin\tfrac{t}{5}\right).$$

Using $y = -\frac{1}{40}x - \frac{1}{2}x'$ gives

$$y = e^{-t/20}\left(-\tfrac{1}{10}B\cos\tfrac{t}{5} + \tfrac{1}{10}A\sin\tfrac{t}{5}\right).$$

From $x(0) = 300$, $A = 300$. From $y(0) = 20$, $-B/10 = 20$, so $B = -200$. Therefore

$$x = e^{-t/20}\left(300\cos\tfrac{t}{5} - 200\sin\tfrac{t}{5}\right), \quad y = e^{-t/20}\left(20\cos\tfrac{t}{5} + 30\sin\tfrac{t}{5}\right).$$

19) d) The vole population is zero when $x = 0$. Since $e^{-t/20} \neq 0$,

$$300\cos\tfrac{t}{5} - 200\sin\tfrac{t}{5} = 0 \quad \Longrightarrow \quad \tan\tfrac{t}{5} = \tfrac{3}{2}.$$

Thus $t = 5\arctan(3/2) = 4.91$ years (3 SF). After this time the model gives negative values for the vole population, so it no longer makes sense.

20) a) Concentration is amount of salt divided by volume of liquid, so it is $x/500$ grams per litre.

20) b) Salt leaves in the 8 litres per minute outflow. Thus the rate is $8(x/500) = 2x/125$ grams per minute.

20) c) The rate in is $4 \times 8 = 32$ g/min and the rate out is $2x/125$ g/min. Therefore

$$\frac{dx}{dt} = \text{rate in} - \text{rate out} = 32 - \frac{2}{125}x.$$

20) d) Salt enters tank 2 from tank 1 at rate $8(x/500) = 2x/125$ g/min. The direct inflow contributes $10 \times 2 = 20$ g/min. Salt leaves at rate $10(y/500) = y/50$ g/min. Hence

$$\frac{dy}{dt} = \frac{2}{125}x + 20 - \frac{1}{50}y.$$

21) a) There is no inflow into tank A. The outflow carries salt at rate $12(x/400) = 3x/100$ g/min, so

$$\frac{dx}{dt} = 0 - \frac{3}{100}x = -\frac{3}{100}x.$$

21) b) The inflow into B carries salt at rate $12(x/400) = 3x/100$ g/min. The outflow carries salt at rate $12(y/400) = 3y/100$ g/min. Thus

$$\frac{dy}{dt} = \frac{3}{100}x - \frac{3}{100}y.$$

21) c) Separating variables in $x' = -3x/100$ and using $x(0) = 60$ gives $x = 60e^{-3t/100}$. Then

$$y' + \tfrac{3}{100}y = \tfrac{9}{5}e^{-3t/100}.$$

The integrating factor is $e^{3t/100}$, so $\frac{d}{dt}(e^{3t/100}y) = 9/5$. Hence $y = \frac{9}{5}te^{-3t/100} + ce^{-3t/100}$. Using $y(0) = 0$ gives $c = 0$, so

$$x = 60e^{-3t/100}, \qquad y = \tfrac{9}{5}te^{-3t/100}.$$

22) a) Tank Q receives liquid from P at 15 L/min and loses liquid back to P at 7 L/min. Without extra outflow, its volume would increase at $15 - 7 = 8$ L/min. To keep the volume constant, liquid must leave Q at 8 L/min.

22) b) The concentrations are $x/700$ in P and $y/700$ in Q. Salt enters P from Q at $7y/700 = y/100$ and leaves P at $15x/700 = 3x/140$. Hence $x' = y/100 - 3x/140$. For Q, salt enters from P at $3x/140$ and leaves at rates $7y/700 = y/100$ and $8y/700 = 2y/175$. Thus

$$y' = \frac{3}{140}x - \left(\frac{1}{100}y + \frac{2}{175}y\right) = \frac{3}{140}x - \frac{3}{140}y.$$

23) a) For tank M, salt enters from the outside at $7c$ g/min and from tank N at $9(y/900) = y/100$ g/min. Salt leaves at $16(x/900) = 4x/225$ g/min. Therefore

$$\frac{dx}{dt} = 7c + \frac{1}{100}y - \frac{4}{225}x = 7c - \frac{4}{225}x + \frac{1}{100}y.$$

23) b) For tank N, salt enters at $16(x/900) = 4x/225$ g/min. It leaves to M at $9(y/900) = y/100$ g/min and to the outside at $7(y/900) = 7y/900$ g/min. Therefore

$$\frac{dy}{dt} = \frac{4}{225}x - \left(\frac{1}{100}y + \frac{7}{900}y\right) = \frac{4}{225}x - \frac{4}{225}y.$$

23) c) At steady state, $x' = 0$ and $y' = 0$. Substituting $x = y = 450$ into the equation for x' gives

$$0 = 7c - \frac{4}{225}(450) + \frac{1}{100}(450), \quad c = \frac{1}{2}.$$

23) d) With $c = 1/2$,

$$x' = \frac{7}{2} - \frac{4}{225}x + \frac{1}{100}y, \qquad y' = \frac{4}{225}x - \frac{4}{225}y.$$

Rearranging the first equation gives $y = 100x' + \frac{16}{9}x - 350$. Differentiating and substituting into the second equation gives

$$x'' + \frac{8}{225}x' + \frac{7}{50625}x = \frac{14}{225}.$$

The auxiliary roots are $-1/225$ and $-7/225$, and a constant particular integral is 450. Thus

$$x = Ae^{-t/225} + Be^{-7t/225} + 450.$$

Using $y = 100x' + \frac{16}{9}x - 350$ gives

$$y = \frac{4}{3}Ae^{-t/225} - \frac{4}{3}Be^{-7t/225} + 450.$$

From $x(0) = 500$ and $y(0) = 400$, $A + B = 50$ and $A - B = -75/2$, so $A = 25/4$, $B = 175/4$. Hence

$$x = \tfrac{25}{4}e^{-t/225} + \tfrac{175}{4}e^{-7t/225} + 450, \quad y = \tfrac{25}{3}e^{-t/225} - \tfrac{175}{3}e^{-7t/225} + 450.$$

24) a) The volume in C remains constant, so total inflow equals total outflow. The outflow from C is 25 L/min and the flow from D to C is 9 L/min. If the external inflow is r L/min, then $r + 9 = 25$, so $r = 16$ L/min.

24) b) For C, salt enters from outside at $16p$ g/min and from D at $9(y/1000) = 0.009y$ g/min. It leaves at $25(x/1000) = 0.025x$ g/min. Thus $x' = 16p - 0.025x + 0.009y$. For D, the printed working uses a 15 L/min flow from D to C and a 10 L/min outlet. These give salt outflow rates $0.015y$ and $0.01y$; salt enters from C at $0.025x$. Hence the printed model is

$$y' = 0.025x - (0.015y + 0.01y) = 0.025x - 0.025y.$$

24) c) With $p = 4$, $x' = 64 - 0.025x + 0.009y$ and $y' = 0.025x - 0.025y$. Rearranging gives $y = \frac{1000}{9}x' + \frac{25}{9}x - \frac{64000}{9}$. Substituting this into the second equation gives

$$x'' + \frac{1}{20}x' + \frac{1}{2500}x = \frac{8}{5}.$$

The auxiliary roots are $-1/100$ and $-1/25$, and the constant particular integral is 4000. Therefore

$$x = Ae^{-t/100} + Be^{-t/25} + 4000, \quad y = \tfrac{5}{3}Ae^{-t/100} - \tfrac{5}{3}Be^{-t/25} + 4000.$$

Using $x(0) = 0$ and $y(0) = 0$ gives $A + B = -4000$ and $A - B = -2400$, so $A = -3200$, $B = -800$. Hence

$$x = -3200e^{-t/100} - 800e^{-t/25} + 4000, \quad y = -\tfrac{16000}{3}e^{-t/100} + \tfrac{4000}{3}e^{-t/25} + 4000.$$

24) d) As $t \to \infty$, both exponential terms tend to zero, so $x \to 4000$ and $y \to 4000$. Each container has 1000 litres, so the limiting concentration in each is $4000/1000 = 4$ g/L.

25) a) From $a' = -3a + 4b + 6$,

$$b = \frac{1}{4}a' + \frac{3}{4}a - \frac{3}{2}, \qquad b' = \frac{1}{4}a'' + \frac{3}{4}a'.$$

Using $b' = -2a + 2b + 10$ gives

$$\frac{1}{4}a'' + \frac{3}{4}a' = -2a + \frac{1}{2}a' + \frac{3}{2}a - 3 + 10, \qquad a'' + a' + 2a = 28.$$

25) b) The auxiliary equation is $m^2 + m + 2 = 0$, so $m = -\frac{1}{2} \pm \frac{\sqrt{7}}{2}i$. A constant particular integral is $a = 14$. Thus

$$a = e^{-t/2}\left(A\cos\frac{\sqrt{7}t}{2} + B\sin\frac{\sqrt{7}t}{2}\right) + 14.$$

25) c) Using $b = \frac{1}{4}a' + \frac{3}{4}a - \frac{3}{2}$ and differentiating the expression for a,

$$b = e^{-t/2}\left[\left(\frac{A}{4} + \frac{\sqrt{7}B}{8}\right)\cos\frac{\sqrt{7}t}{2} + \left(\frac{B}{4} - \frac{\sqrt{7}A}{8}\right)\sin\frac{\sqrt{7}t}{2}\right] + 9.$$

25) d) Since $a(0) = 4$, $A = -10$. Since $b(0) = 1$,

$$1 = \frac{A}{4} + \frac{\sqrt{7}B}{8} + 9, \quad B = -2\sqrt{7}.$$

Hence

$$a = e^{-t/2}\left(-10\cos\frac{\sqrt{7}t}{2} - 2\sqrt{7}\sin\frac{\sqrt{7}t}{2}\right) + 14, \qquad b = -8e^{-t/2}\cos\frac{\sqrt{7}t}{2} + 9.$$

25) e) As $t \to \infty$, the exponential terms tend to zero, so the limiting values are $a = 14$ and $b = 9$.

25) f) At $t = 6$, the solutions give $a \approx 13.7788$ and $b \approx 9.0331$. They are close to their limiting values. The approach is oscillatory: at $t = 7$, $a \approx 14.2716$ and $b \approx 9.2383$, slightly farther from equilibrium than at $t = 6$.

25) g) At equilibrium, $a' = b' = 0$, so

$$3a - 4b = 6, \qquad 2a - 2b = 10,$$

which gives $(a, b) = (14, 9)$. The constant terms shift the equilibrium; the other coefficients determine the approach to it.

26) a) From $p' = 2p - 2r - 341e^{-t}$,

$$r = -\frac{1}{2}p' + p - \frac{341}{2}e^{-t}, \qquad r' = -\frac{1}{2}p'' + p' + \frac{341}{2}e^{-t}.$$

Using $r' = 4p - 4r$ gives $p'' + 2p' = -1023e^{-t}$.

26) b) The auxiliary roots are $0, -2$, so $p_c = A + Be^{-2t}$. Trying $p_p = ke^{-t}$ gives $-ke^{-t} = -1023e^{-t}$, hence

$$p = A + Be^{-2t} + 1023e^{-t}.$$

26) c) Substitution into the expression for r gives

$$r = A + 2Be^{-2t} + 1364e^{-t}.$$

26) d) From $p(0) = 2046$ and $r(0) = 3411$,

$$A + B = 1023, \qquad A + 2B = 2047,$$

so $A = -1$, $B = 1024$. Therefore

$$p = -1 + 1024e^{-2t} + 1023e^{-t}, \qquad r = -1 + 2048e^{-2t} + 1364e^{-t}.$$

26) e) Setting $p = 0$ at $t = T$ gives

$$e^{2T} - 1023e^{T} - 1024 = 0 = (e^{T} + 1)(e^{T} - 1024).$$

Since $e^T > 0$, $T = \ln 1024 \approx 6.931$ (3 d.p.).

26) f) As $t \to \infty$, $r \to -1$. This is not realistic for a population, so the model eventually breaks down.

27) a) From $x' = a(x - y)$, $y = x - x'/a$, and hence $y' = x' - x''/a$. Substituting into $y' = ay - bx$ gives

$$x'' - 2ax' + a(a - b)x = 0.$$

27) b) The auxiliary equation is $m^2 - 2am + a(a - b) = 0$, so

$$(m - a)^2 = ab, \qquad m = a \pm \sqrt{ab},$$

and

$$x = Ae^{(a+\sqrt{ab})t} + Be^{(a-\sqrt{ab})t}.$$

27) c) Using $y = x - x'/a$ gives

$$y = -A\sqrt{\frac{b}{a}}e^{(a+\sqrt{ab})t} + B\sqrt{\frac{b}{a}}e^{(a-\sqrt{ab})t}.$$

27) d) Substitution yields

$$x + \sqrt{\frac{a}{b}}\,y = 2Be^{(a-\sqrt{ab})t}, \qquad x - \sqrt{\frac{a}{b}}\,y = 2Ae^{(a+\sqrt{ab})t}.$$

Thus the constants in the stated forms are $K_1 = 2B$ and $K_2 = 2A$.

27) e) For $A \neq 0$, the larger exponential dominates as $t \to \infty$, so

$$\frac{y(t)}{x(t)} \longrightarrow -\sqrt{\frac{b}{a}}.$$

27) f) If $A = 0$ (equivalently $K_2 = 0$), the other exponential dominates and $y(t)/x(t) \to \sqrt{b/a}$. Therefore the claim does not hold in this case.

28) a) If A decreases proportionally to its amount, $a' = -pa$ for a positive constant p. B is gained from A and lost by conversion to C, giving $b' = pa - qb$, where $q > 0$.

28) b) Since no material is added or lost, $a + b + c$ is constant. Therefore

$$c' = -a' - b',$$

so the third differential equation is determined by the first two and is not independent.

29) a) Separating variables and using $p(0) = 10$,

$$\int \frac{1}{p}\,dp = \int -2\,dt, \quad \ln|p| = -2t + C, \quad p = 10e^{-2t}.$$

29) d) Since no material is lost, $p + q + r = k$ and

$$\frac{d}{dt}(p + q + r) = 0, \qquad p' + q' + r' = 0.$$

29) c) The total amount of material remains constant: material transfers from P to Q and then to R, with none created or lost.

29) d) From $q' = 2p - q$ and $p = 10e^{-2t}$,

$$q' + q = 20e^{-2t}.$$

The integrating factor is e^t, so $(e^t q)' = 20e^{-t}$ and

$$e^t q = -20e^{-t} + C, \qquad q = -20e^{-2t} + Ce^{-t}.$$

Using $q(0) = 0$ gives $C = 20$, hence $q = 20e^{-t} - 20e^{-2t}$.

29) e) As $p + q + r = 10$,

$$r = 10 - p - q = 10 + 10e^{-2t} - 20e^{-t}.$$

30) a) Separating variables and using $x(0) = 100$,

$$\int \frac{1}{x}\,dx = \int -\frac{3}{100}\,dt, \quad \ln|x| = -\frac{3t}{100} + C, \quad x = 100e^{-3t/100}.$$

30) b) From $y' = \frac{3}{100}x - \frac{1}{100}y$, $x = \frac{100}{3}y' + \frac{1}{3}y$. Differentiating and using $x' = -\frac{3}{100}x$ gives

$$y'' = -\frac{9}{10000}x - \frac{1}{100}y' = -\frac{3}{100}y' - \frac{3}{10000}y - \frac{1}{100}y',$$

so $y'' + \frac{1}{25}y' + \frac{3}{10000}y = 0$.

30) c) The auxiliary roots are $-1/100$ and $-3/100$, so $y = Be^{-t/100} + Ce^{-3t/100}$. Since $y(0) = 0$, $B + C = 0$. The original equation gives $y'(0) = 3$, hence

$$-\frac{B}{100} - \frac{3C}{100} = 3, \qquad B + 3C = -300.$$

Thus $B = 150$, $C = -150$, and $y = 150e^{-t/100} - 150e^{-3t/100}$.

30) d) Since $x + y + z = 100$,

$$z = 100 - x - y = 100 - 150e^{-t/100} + 50e^{-3t/100}.$$

As $t \to \infty$, $z \to 100$ kg.